Alejandro Scaffa, PhD

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Profile

Scientist with a Ph.D. in Molecular Pharmacology and Physiology from Brown University and postdoctoral training at Merck Research Labs. Currently completing an M.S. in Computer Science. Ph.D. research investigated the long-term impact of supplemental oxygen exposure on premature lungs, uncovering links to cellular senescence and metabolic dysfunction. Postdoctoral work at Merck focused on the tumor microenvironment, cancer-associated fibroblasts, and senescence. Additional experience in venture capital through a rotation at the MRL Ventures Fund. Fast learner, effective communicator, and highly collaborative team member.

Education

Northeastern University (Roux Institute), Portland, Maine, USA
M.S. Computer Science (Align Track) — Sept 2023 - May 2026 (expected)

Brown University, Providence, Rhode Island, USA
Ph.D. Molecular Pharmacology and Physiology — May 2020
Mentor: Phyllis Dennery M.D.

M.A. Molecular Pharmacology and Physiology — May 2016

Grinnell College, Grinnell, Iowa, USA — May 2014
B.A. with honors, Biochemistry

Danish Institute for Study Abroad (DIS), Copenhagen, Denmark — Aug - Dec 2012
Visiting Student, Biomedicine

Professional Experience

Postdoctoral Fellow, Merck Research Laboratories, Boston, MA — Sept 2020 - May 2023
Postdoc in tumor microenvironment leading innovative research in fibroblasts and senescence within pancreatic ductal adenocarcinoma (PDAC). Including multidisciplinary collaborations in biology, chemistry, and bioinformatics. Other roles include the use of innovative techniques like scRNASeq, RNAseq, secretomics, and bioinformatics analysis, as well as project management.

Intern (rotation), Merck Research Laboratories Ventures, Boston, MA — Oct 2021 - Jul 2022
The MRL Ventures Fund (MRLV) team invests in early-stage, preclinical therapeutics companies. This associate level internship position includes meeting with venture creators, early diligence, landscape analysis, investing, and large picture strategy.

Postdoctoral Researcher, Brown University, Providence, RI — Jun 2020 - Aug 2020
Postdoc at the laboratory of Phyllis Dennery M.D. Continuing research from PhD described below.

Skills

Programming Languages: Python, Java, and C. Basic knowledge of R and C++.

Research: Cancer, genomics, senescence, literature review, project design and management, CRISPR, cell culture, metabolism, (single cell) RNA-sequencing, flow cytometry, Seahorse bioanalyzer, Nanostring, and western blots. I rely on problem solving, communication, and innovative techniques.

Languages: Fluent: English, Portuguese, and Spanish.

PhD Research

Hyperoxia (Supplemental oxygen) leads to Senescence and Glycolytic Dysregulation in ATII lung cells, a model for Bronchopulmonary Dysplasia - P.I.: Prof. Dennery MD
PhD focused on negative sequelae from life-saving lung ventilation with supplemental oxygen (hyperoxia) given to premature babies. Described mechanisms leading to hyperoxia-induced senescence in vitro and in vivo, and compositional changes to the lung after hyperoxia through scRNAseq and innovative imaging analysis.

Honors and Awards

Tarr-Coyne Family Scholarship, The Roux Institute of Northeastern University 2025

Bertelsmann Tech. Chall. Scholarship, Machine Learning, Bertelsmann and Udacity 2021

Google Developer Challenge Scholarship, Google and Udacity 2018

Best Aging in Place Hack, MIT Hacking Medicine Grand Hack, Cambridge, MA 2016

Pharmacology Pre-Doctoral Fellow, Brown University, Providence, RI 2015

1st Prize in Biochemistry, Best in Category, and naming rights of a minor planet 2010
Intel® International Science and Engineering Fair (ISEF) co-hosted by Google, San Jose, CA

Publications, Presentations, and Abstracts

Scaffa A, Hornburg M, Kopsiaftis S. Single-Cell Transcriptomics of Senescent Cancer-Associated Fibroblasts. 2024; [unpublished - industry research].

Yao H, Wallace J, Peterson A, **Scaffa A**, Rizal S, Hegarty K, Maeda H, Chang J, Oulhen N, Kreiling J, Huntington K, De Paepe M, Barbosa G, Dennery PA. Timing and cell specificity of senescence drives postnatal lung development and injury. Nature Commun 14, 273. 2023.

Scaffa A, Tollefson G, Yao, H, Rizal S, Wallace J, Oulhen N, Carr J, Hegarty K, Uzun A, Dennery PA. Identification of Heme Oxygenase-1 as a Putative DNA-Binding Protein. Antioxidants. 2022 Nov; 11:2135. DOI: 10.3390/antiox11112135.

Scaffa A, Yao, H, Oulhen, N, Dennery, PA. Single-cell transcriptomics reveals lasting changes in the lung cellular landscape into adulthood after neonatal hyperoxic exposure. Redox Biology. 2021 Dec; 48:102091. PMID: 34417156.

Scaffa AM, Peterson AL, Carr JF, Garcia D, Yao H, Dennery PA. Hyperoxia causes senescence and increases glycolysis in [...] epithelial cells. Physiol Rep. 2021 May; 9(10):e14839. PMID: 34042288.

Garcia D, Carr JF, Chan F, Peterson AL, Ellis KA, **Scaffa A**, Ghio AJ, Yao H, Dennery PA. Short exposure to hyperoxia causes cultured lung epithelial cell mitochondrial dysregulation and alveolar simplification in mice. Pediatr Res. 2021 Jul; 90(1):58-65. PMID: 33144707.

Carr JF, Garcia D, **Scaffa A**, Peterson AL, Ghio AJ, Dennery PA. Heme Oxygenase-1 Supports Mitochondrial Energy Production and Electron Transport Chain Activity in Cultured Lung Epithelial Cells. Int J Mol Sci. 2020 Sep; 21(18):6941. PMID: 32971746.

Scaffa A, Peterson, A, Dennery PA. Hyperoxic Exposure Increases Senescence [...] Role Of p53 Signaling. Am J Respir Crit Care Med. A61 Epithelial Biology. 2019; 199:A7327. doi:10.1164.

Scaffa A, Peterson, A, Dennery PA. Hyperoxic exposure causes senescence in [...] lung epithelial cells. Pediatric Academic Society. 3700 Pulmonology: Biology, Epidem., and Phys. 2019; 3700.6.

Scaffa A, Glidden, D, Soemedi R, Fairbrother W. Engineering splicing mutations in HEK 293 cells using CRISPR/Cas9 system. The CRISPR/Cas Revolution, CSHL. 2015 Oct; 127.

Teaching Experience (Abbreviated)

Brazilian Portuguese Curriculum Developer, Grinnell College Jan 2013 - May 2014
Originally an instructor, developed a new syllabus and course for Portuguese I and II.

Biochemistry Teaching Assistant, Grinnell College Aug - Dec 2013
Attended all of Prof. Trimmer's classes, held office hours, mentor sessions, and other needed duties.

Community Involvement (Abbreviated)

Lead, ALIANZA EBRG (Boston/Cambridge), Merck Mar 2023 - May 2023

President of SACNAS, Brown University Jul 2018 - Jun 2019

Member, Biomedical Science Career Program, Harvard Medical School Jun 2016 - Present
Active member, panelist, abstract judge for NESS, and mentor to high school students.

Certifications

Python for Data Science and AI, Coursera, 2019. **Data Science Methodology**, Coursera, 2019. **Open Source Tools for Data Science**, Coursera, 2019. **A Crash Course in Data Science**, Coursera, 2018. **Teaching Certificate I**, Sheridan Center. Brown University, 2016.